F78PB Probability and Statistics B
Tutorial 2: Statistical models and properties of estimators

1. Which of the following statements relate to probabilities that can be interpreted as frequencies? For those that can, describe the corresponding sequence of experiments.

   (a) The probability that a '6' is scored with a given die is $\frac{1}{6}$.

   (b) The probability that a randomly chosen 20-year old male survives until age 60 is 0.8.

   (c) The probability that George Clooney survives until age 70 is 0.8.

   (d) The probability that Jack the Ripper was a member of the Royal Family is 0.05.

   (e) The probability that the next pint of milk you buy goes off within 1 week of purchase is 0.6.

2. Which standard distributions might be appropriate to model the following experimental outcomes?

   (a) Select 10 individuals randomly from the student population of Heriot-Watt and count the number of 1st-year students in your sample.

   (b) Throw a dart at a dartboard repeatedly until you score a treble 20. Count the number of throws required.

   (c) Monitor a stretch of road continuously for 1 year and count the number of accidents that occur.

   (d) Select a male student at random from the HW population and measure their height.

   (e) Put a new battery in a watch and measure the time elapsed before the watch stops.

3. Suppose that $X_1, X_2, ..., X_n$ are *independent* and *identically distributed* random variables, each with a Gamma (t, λ) distribution, and let $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Determine $M_{\bar{X}}(\theta)$, the moment generating function of $\bar{X}$ and hence, or otherwise, determine the sampling distribution of $\bar{X}$.

4. Let $\underline{X}$ denote a random sample of size n from a distibution with parameter θ and let $g(\underline{X})$ denote an estimator for θ. Recall that

$$\text{Bias}(g(\underline{X})) = \mathbb{E}[g(\underline{X})] - \theta \quad \text{and} \quad \text{MSE}(g(\underline{X})) = \mathbb{E}[(g(\underline{X}) - \theta)^2].$$

   Show that $\text{MSE}(g(\underline{X})) = \mathbb{V}\text{ar}(g(\underline{X})) + (\text{Bias}(g(\underline{X})))^2$.

5. Suppose that $X_1, X_2, ..., X_n$ are i.i.d. random variables with mean μ and variance σ^2. Let $\bar{X}$ and S^2 denote the sample mean and variance (calculated from the random sample $X_1, X_2, ..., X_n$). Show that

   (a) $\mathbb{E}(\bar{X}) = \mu$, (b) $\mathbb{V}\text{ar}(\bar{X}) = \frac{\sigma^2}{n}$, (c) $\mathbb{E}(S^2) = \sigma^2$.

   Hint: In (c) expand the sum of squares in S^2. Also recall that
   $\mathbb{V}\text{ar}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$ and similarly $\mathbb{V}\text{ar}(\bar{X}) = \mathbb{E}(\bar{X}^2) - (\mathbb{E}(\bar{X}))^2$.

6. Let $\underline{X}$ denote a random sample of size n from a Poi(λ) distribution. Consider the estimator $\hat{\lambda}(\underline{X}) = \bar{X}$.

   (a) Is $\hat{\lambda}(\underline{X})$ an unbiased estimator for λ? Give reasons.

   (b) Calculate the MSE($\hat{\lambda}(\underline{X})$).

   (c) Consider the 'family' of estimators $\hat{\lambda}(\underline{X}) = a\bar{X}$ where $a > 0$. Calculate the MSE of such an estimator as a function of a. Which value of a gives the estimator with the minimum MSE? Comment on your answer. What does the value of a tend to as the sample size n grows to infinity?

7. Let $\underline{X}$ be a random sample from $\mathrm{N}(0, \sigma^2)$. Find the bias and comment on the efficiency of the following estimators for σ^2.

$$\begin{aligned} S^2 &= \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \\ S_n^2 &= \frac{1}{n}\sum_{i=1}^{n} X_i^2 \\ S_{n+2}^2 &= \frac{1}{n+2}\sum_{i=1}^{n} X_i^2. \end{aligned}$$

   Note: You can use the fact that for a random sample from $\mathrm{N}(\mu, \sigma^2)$, we have $\mathbb{V}\mathrm{ar}(S^2) = \frac{2\sigma^4}{n-1}$. Also, for a $\mathrm{N}(0, \sigma^2)$ random variable X, $\mathbb{V}\mathrm{ar}(X^2) = 2\sigma^4$.
   Also, for the last equation, write S_{n+2}^2 in terms of S_n^2 and use the results obtained for S_n^2.

8. Suppose that $\underline{X} = (X_1, X_2, ..., X_n)$ is a random sample from a distribution with density function

$$f_X(x;\theta) = \begin{cases} \frac{2x}{\theta^2} & \text{if} \quad 0 \le x \le \theta, \\ 0 & \text{if } x < 0 \text{ or } x > \theta. \end{cases}$$

   (a) Show that the estimator

$$\hat{\theta}(\underline{X}) = \frac{3\bar{X}}{2},$$

   where $\bar{X}$ denotes the sample mean, is an unbiased estimator for the parameter θ and calculate MSE($\hat{\theta}(\underline{X})$).

   (b) Consider the family of estimators for θ of the form $Y = a\bar{X}$. Express the MSE of $a\bar{X}$ as function of a and identify the value of a^* which defines the *most efficient* estimator of this form.

9. Let $\underline{X}$ be a random sample from Exp($1/\theta$) (i.e. with mean θ).

   (a) Write down $\mathbb{E}(\bar{X}), \mathbb{V}\mathrm{ar}(\bar{X})$ and MSE($\bar{X}$).

(b) Now consider the estimator $Y = a\bar{X}$. Write down MSE(Y) and determine the value of a which minimizes it. Compare $\bar{X}$ and the optimum Y for both small and large values of n. Comment. What does the value optimal value of a tend to as n grows to infinity?

10. For a random sample from $\mathrm{N}(\mu, \sigma^2)$, consider the estimator

$$Y = a\sum_{i=1}^{n}(X_i - \bar{X})^2 = a(n-1)S^2$$

for σ^2. Show that $\mathrm{MSE}(Y) = 2a^2(n-1)\sigma^4 + (a(n-1)-1)^2\sigma^4$ and show that MSE(Y) is minimised when $a = 1/(n+1)$. Comment on your result.

11. A gambler is concerned that a coin may not be fair and conducts an experiment to estimate the probability $p \in (0,1)$ of obtaining a head (H). He tosses the coin 8 times, and he considers the outcome of this experiment as a random sample $\underline{X} = (X_1, X_2, ..., X_8)$ of size 8 from a Ber(p) distribution where $X_i = 1$ if the ith toss results in 'H' and is 0 otherwise.

Consider two estimators for p defined by $g_1(\underline{X}) = \frac{Y}{8}$ and $g_2(\underline{X}) = \frac{Y+1}{10}$ where $Y = \sum_{i=1}^{8} X_i \sim \mathrm{Bin}(8, p)$.

(a) Determine MSE($g_1(\underline{X})$) and MSE($g_2(\underline{X})$) as *functions of p*.

(b) Sketch the graphs of MSE($g_1(\underline{X})$) and MSE($g_2(\underline{X})$) (as *functions of p*), and determine the values of p for which $g_2(\underline{X})$ is more efficient than $g_1(\underline{X})$.

Exam Style Question

(a) Suppose that $\underline{X} = (X_1, X_2, \ldots, X_n)$ is a random sample from a distribution parametrised by unknown θ and let $g(\underline{X})$ denote an estimator for θ.

(i) Give definitions for the *bias* and *mean square error* (MSE) of $g(\underline{X})$.

[2 marks]

(ii) Express the mean square error in terms of the variance, $\mathbb{V}\mathrm{ar}(g(\underline{X}))$, and bias of $g(\underline{X})$.

[1 marks]

(b) Let $\underline{X}$ be a random sample of size n from a distribution with density function

$$f_X(x;\theta) = \begin{cases} \frac{1}{\theta^2} x e^{-\frac{x}{\theta}}, & x > 0 \\ 0, & x \leq 0. \end{cases}$$

Here $\theta > 0$ is an unknown parameter. Note that this corresponds to a Gamma $\left(2, \frac{1}{\theta}\right)$ distribution and has mean 2θ and variance $2\theta^2$.

(i) Consider $\hat{\theta}_1(\underline{X}) = \frac{1}{2}\bar{X}$ as an estimator of θ. Show that

$$\text{MSE}(\hat{\theta}_1(\underline{X})) = \frac{\theta^2}{2n}$$

and explain what happens to the mean square error as n is increased.

[4 marks]

(ii) Now consider $\hat{\theta}_2(\underline{X}) = c\bar{X}$ as an estimator of θ where $c > 0$ is a constant. Show that

$$\text{MSE}(\hat{\theta}_2(\underline{X})) = \frac{2c^2\theta^2}{n} + (2c-1)^2\theta^2.$$

[4 marks]

(iii) Determine the value of c that minimises the mean square error of $\hat{\theta}_2$.

[3 marks]

[Total 14 marks]